BIS SCIENCE PROJECTS 2026 — CLASS 7 (EV)

EARTHQUAKE & TREMOR ALARM SYSTEM

An IoT-Enabled Hardware-Interrupt Early Warning System

System Architecture & Hardware Specification Blueprint

PROJECT TEAM ROSTER & ROLE ALLOCATIONS

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PROJECT LEAD • SYSTEM ARCHITECT • CODER • BUILDER • PLANNER

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ACADEMIC & PROJECT OVERVIEW

Academic Event	BIS Science Projects 2026 / Class 7 English Version (EV)
Project Author & Lead	**Abdullah Karim** (System Architect & Lead Engineer)
Project Objective	Design and build an earthquake early warning system that triggers visual/acoustic alarms locally and sends remote alerts over Wi-Fi within 15ms.

COMPLETE BILL OF MATERIALS (BOM) & COMPONENT SOURCING

#	COMPONENT	TECHNICAL SPECIFICATION	QTY	USD COST	BDT COST
1	**ESP32 Board**	ESP-WROOM-32 (30-Pin NodeMCU, Dual Core, CP2102)	1	$0.00 (Owned)	৳0.00 (Owned)
2	**SW-420 Sensor**	Vibration Sensor Module with LM393 Comparator	1	$1.20	৳145.00
3	**Active Buzzer**	5V/3.3V Sound Transducer, Sound Level >85dB	1	$0.40	৳50.00
4	**BC547 NPN**	TO-92 NPN Transistor (45V 100mA Switch)	1	$0.10	৳12.00
5	**1 kΩ Resistor**	1/4W Metal Film, ±5% (Transistor Base Protection)	1	$0.02	৳3.00
6	**330 Ω Resistor**	1/4W Metal Film, ±5% (LED Current Limiter)	1	$0.02	৳3.00
7	**Red LED (5mm)**	High-brightness Diffused Red LED (Vf = 2.0V)	1	$0.08	৳10.00
8	**Breadboard**	MB-102 830-Point Solderless Breadboard	1	$1.50	৳180.00
9	**Jumper Wires**	20cm Male-to-Male Jumpers Set	10	$0.38	৳45.00
		TOTAL OUT-OF-POCKET COST:	**17**	**$3.70 USD**	**৳448.00 BDT**

ABDULLAH KARIM

EARTHQUAKE & TREMOR ALARM SYSTEM

WIRING SPECIFICATION & ASSEMBLY WALKTHROUGH

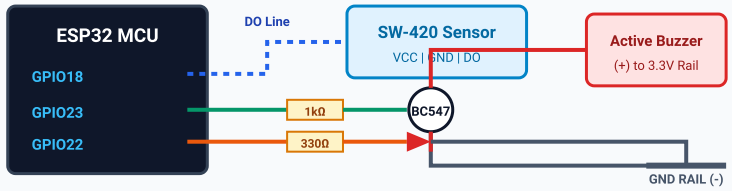
Doc ID: ETA-2026-SPEC-V5
Revision: 5.0.0 (Final)

TRIGGER RESPONSE	OPERATING VOLTAGE	ACOUSTIC OUTPUT	ADJUSTED TOTAL COST
< 15 ms	3.3V DC	85 dBA	$3.70 USD
Hardware Interrupt	5V USB Input	@ 10cm Distance	~448 BDT (ESP32 Owned)

HARDWARE MASTER WIRING SPECIFICATION MATRIX

PIN ID	WIRE COLOR	DESTINATION COMPONENT	LOGIC / RAIL	BOARD COORDINATES
GPIO 18	Blue Cable	SW-420 Signal Pin (DO)	3.3V DIGITAL IN	ESP Pin 18 → Row 23 Pin DO
GPIO 23	Green Cable	1kΩ Resistor → BC547 Base	3.3V DIGITAL OUT	ESP Pin 23 → Row 27 (Base)
GPIO 22	Orange Cable	330Ω Resistor → LED Anode	3.3V DIGITAL OUT	ESP Pin 22 → Row 30 (Anode)
3V3	Red Cable	Breadboard Power Rail (+)	+3.3V DC RAIL	ESP 3V3 Pin → Bus Rail (+)
GND	Black Cable	Breadboard Ground Rail (-)	0V GROUND RAIL	ESP GND Pin → Bus Rail (-)

SYSTEM SCHEMATIC ARCHITECTURE (VECTOR DIAGRAM)



COMPREHENSIVE 8-STEP PHYSICAL CONSTRUCTION GUIDE

1 Mount ESP32 Module
- Place ESP32 across central divide ditch (Rows 1–15).
- Ensure Pin 3V3 sits on Row 1 (Left) and Vin on Row 1 (Right).

2 Power Rails setup
- Connect jumper wire from ESP32 **3V3** to Red (+) Rail.
- Connect jumper wire from ESP32 **GND** to Black (-) Rail.

3 Position Sensor
- Mount SW-420 sensor vertically on breadboard Rows 20–23.
- Jumper SW-420 **VCC** to 3.3V rail and **GND** to ground rail.

4 Sensor Calibration
- Connect SW-420 **DO** pin to ESP32 **GPIO 18**.
- Turn blue potentiometer counter-clockwise to adjust trigger limit.

5 Install Transistor
- Insert BC547 on Rows 26–28; wire **1kΩ resistor** from **GPIO 23** to Middle Pin.
- Connect Right Pin (Emitter) to ground rail (-).

6 Connect Buzzer
- Connect buzzer (+) anode to 3.3V power rail (+).
- Connect (-) cathode to BC547 Left Pin (Collector).

7 Status LED Assembly
- Bridge **330Ω resistor** from **GPIO 22** to LED anode Row 30.
- Connect LED cathode directly to ground rail (-).

8 Power & Verification
- Inspect wiring. Plug USB to ESP32; confirm power LED glows.
- Tap sensor to test immediate local buzzer & LED alert.

Power Supply & Protection Note: The system is powered by a standard 5V USB source (phone charger or power bank) plugged into the ESP32. Transistor switching isolates piezoceramic voltage spikes from ruining MCU GPIO digital pins.

ABDULLAH KARIM

FIRMWARE ARCHITECTURE & REMOTE ALERT DISPATCH

HARDWARE INTERRUPT & NETWORK GATEWAYS

Hardware ISR & Software Backends

CORE HARDWARE INTERRUPT LOGIC (IRAM_ATTR ISR)

```
// Optimized Hardware ISR Logic Implementation — Abdullah Karim #include <Arduino.h> #define SENSOR_PIN 18 #define BUZZER_PIN 23 #define LED_PIN 22 volatile bool tremorDetected = false; unsigned long lastTriggerTime = 0; void IRAM_ATTR sensorISR() { unsigned long currentTime = millis(); if (currentTime - lastTriggerTime > 500) { // 500ms debounce tremorDetected = true; lastTriggerTime = currentTime; } } void setup() { pinMode(SENSOR_PIN, INPUT_PULLUP); pinMode(BUZZER_PIN, OUTPUT); pinMode(LED_PIN, OUTPUT); attachInterrupt(digitalPinToInterrupt(SENSOR_PIN), sensorISR, FALLING); }
```

COMPREHENSIVE REMOTE NOTIFICATION GATEWAYS (5 BACKENDS)

1. Telegram HTTPS Bot — SSL REST

```
#include <WiFiClientSecure.h> void sendTelegramAlert(String token, String chat) { WiFiClientSecure client; client.setInsecure(); if (client.connect("api.telegram.org", 443)) { client.print("GET /bot" + token + "/sendMessage?chat_id=" + chat + "&text=TREMOR_ALERT HTTP/1.1\r\nHost: api.telegram.org\r\n\r\n"); } }
```

2. WhatsApp CallMeBot — REST API

```
#include <HTTPClient.h> void sendWhatsAppAlert(String phone, String key) { HTTPClient http; http.begin("https://api.callmebot.com/whatsapp.php?phone=" + phone + "&text=TREMOR_DETECTED&apikey=" + key); http.GET(); http.end(); }
```

3. Twilio Cellular SMS — BASIC AUTH

```
HTTPClient http; http.begin("https://api.twilio.com/2010-04-01/Accounts/" + sid + "/Messages.json"); http.setAuthorization(sid.c_str(), token.c_str()); http.POST("To=" + targetNum + "&From=" + twilioNum + "&Body=EARTHQUAKE_ALERT");
```

4. Async WebSockets — < 15MS LOCAL

```
#include <ESPAsyncWebServer.h> AsyncWebSocket ws("/ws"); void triggerLocalBroadcast() { if (ws.count() > 0) ws.textAll("{\"status\":\"CRITICAL_TREMOR\"}"); }
```

5. Encrypted SMTP Email Gateway — TLS PORT 465

```
#include <ESP_Mail_Client.h> SMTPSession smtp; Session_Config config; config.server.host_name = "smtp.gmail.com"; config.server.port = 465; config.login.email = SENDER_EMAIL; config.login.password = APP_PASSWORD; // Connects via SSL and dispatches detailed event telemetry report
```

FLASHING & DEPLOYMENT MANUAL

1. Open Arduino IDE (v2.0+) and add ESP32 URL in Preferences.
2. Install ESP32 Board Manager package and select **ESP32 Dev Module**.
3. Install `ArduinoJson`, `ESPAsyncWebServer`, and `ESP-Mail-Client` libraries.
4. Set Upload Speed to `921600` and CPU Frequency to `240MHz`. Flash firmware.